Machine Learning: Simple Neural Networks

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The brain as model

« There is a fantastic existence proof that learning is possible, which is the bag of water and electricity (together with a few trace chemicals) sitting between your ears […] wich is the squishy thing that your skull protects »

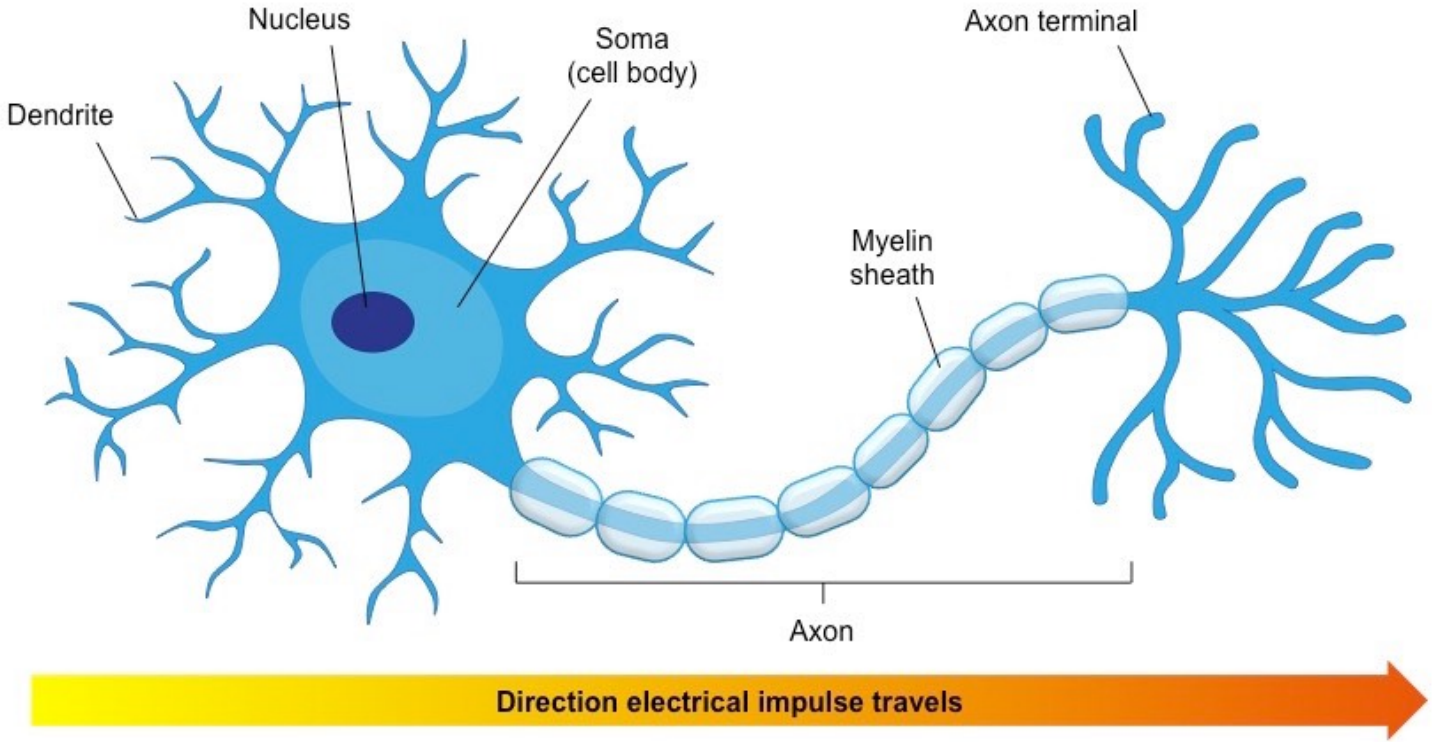
**Stephen Marsland*

The brain does exactly what we want for data analysis:

- Extract complex information in a compressed form
- Deal with noisy and/or inconsistent data
- Work in highly-dimensional spaces
- Give the appropriate answer most of the time
- Provide results very quickly
- Remain robust through aging (neuron loss)

The biological neuron

Elementary brick of a biological brain (10^{11} in the human brain).
The idea will be to use it as a model to emulate learning capabilities.



Perform the sum of various electrochemical input signals. If this total signal is sufficient, it sends a new signal through its axon to transfer information (toward other neurons).

Biological Neural Networks

A connection between two neurons is called a synapse (10^{14} in the human brain).
A neuron is a binary compute unit that either “fire” or “not-fire” in response to a signal.

➡ **In this view, a brain is a massively parallel super-computer of 10^{11} processing units.**

A simplified view of how it learns

The synapses represent the “strength” of the connection between two neurons.

Learning = modifying this connection (either positive or negative and changing its intensity)

➡ **This is called plasticity**

The Hebb law defines a simplified rule for learning:

If two neurons “fire” at the same time, there must be some **correlation** between them, and their connection must be strengthened.

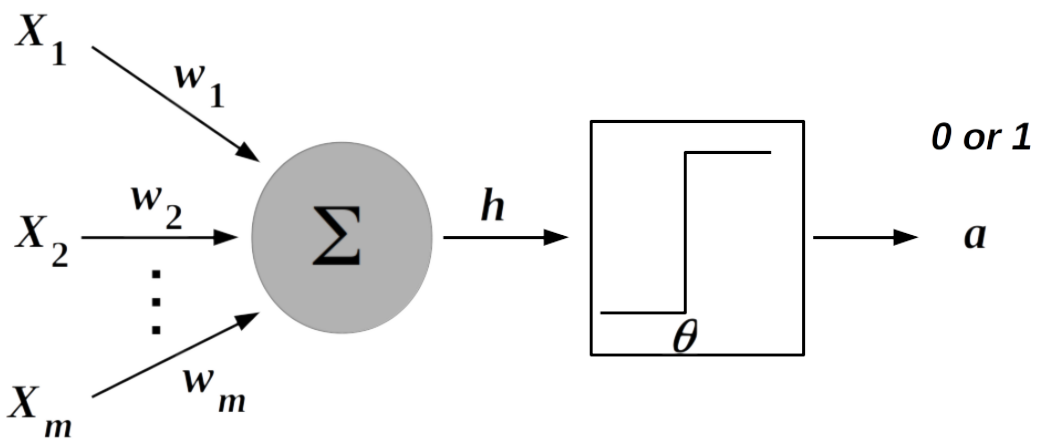
➡ **This is called conditioning**

**These rules are not enough to train a neural system but illustrate the processes that occur in the biological brain.*

To create an algorithm from these biological concepts, one first need to create a mathematical model.

Model of a Neuron

Mathematical model from McCulloch and Pitts, inspired by the biological neuron.

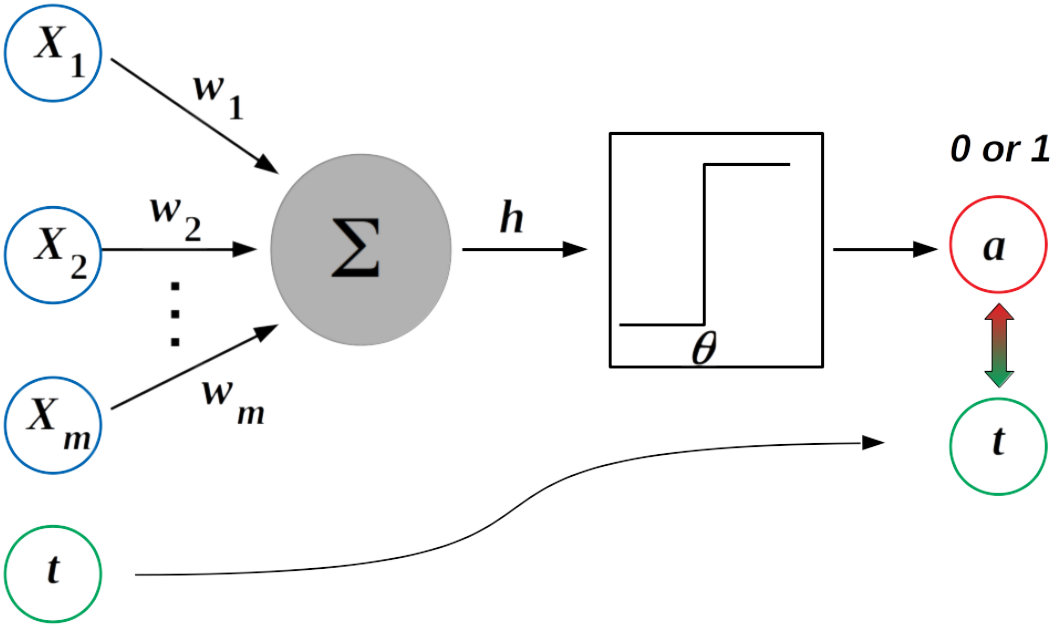


$$h = \sum_{i=1}^{m} X_i w_i \qquad a = g(h) = \begin{cases} 1 & \text{if} \quad h > \theta \\ 0 & \text{if} \quad h \leq \theta \end{cases}$$

Its main components are:

- An input vector X_i that represents the dimensions of a given object
- A set of weights w_i that links the various input dimensions to the neuron
- A sum function h that defines how these weights are combined with the input dimensions
- An activation function $g(h)$ that defines if the network should remain in a “0” state or should be activated to a “1” state, depending on the results of the sum.

Training a Neuron



$$h = \sum_{i=1}^{m} X_i w_i \qquad a = g(h) = \begin{cases} 1 & \text{if} \quad h > \theta \\ 0 & \text{if} \quad h \leq \theta \end{cases}$$

The learning process for this model is supervised
→ Each input vector is associated to a target value

So what is the purpose of the neuron if the expected value is already known ?

➡ **Generalization**

Find patterns in the data distribution in the parameter space so it can perform prediction on vectors with unseen (but close) values.

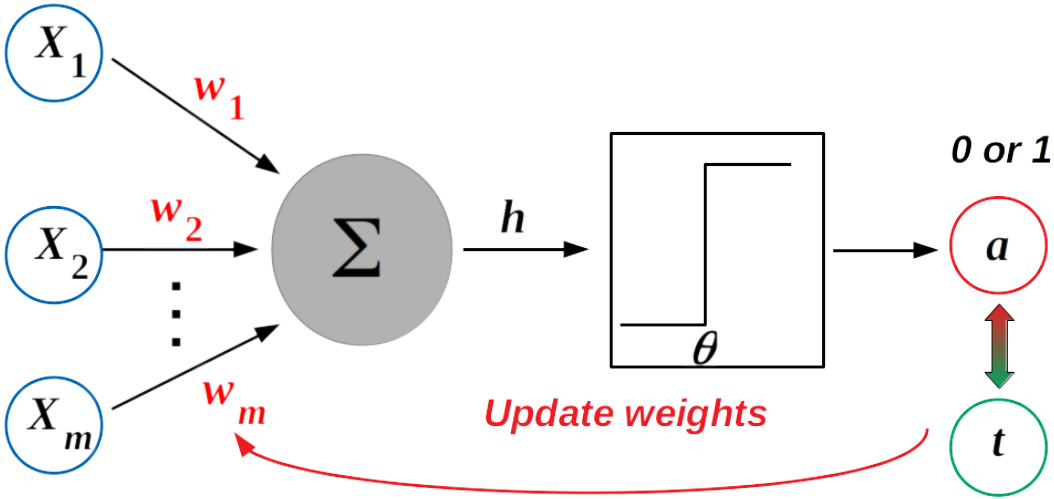
In practice, such a simple neuron optimizes a linear separation in the parameter space (**hyperplane**).

How does this model learn ?

Which parameters can or cannot be modified ?

The input, output, and targets are fixed, so the learning relies on modifying the **weights and the activation threshold**.

Training a Neuron



$$\omega_i \leftarrow \omega_i - \eta\ (a - t)\ x_i$$

$$h = \sum_{i=1}^{m} X_i w_i \qquad a = g(h) = \begin{cases} 1 & \text{if} \quad h > \theta \\ 0 & \text{if} \quad h \leq \theta \end{cases}$$

How to proceed when the output does not correspond to the target ?

There are *m* weights w_i associated with the network corresponding to the input dimensions.

How to modify the weights?

- If the output state is 1 while it should be 0, the weights need to be lowered
- If the output state is 0 while it should be 1, the weights need to be increased

To quantify this modification, one needs to choose an error function *E.*

$$E = 0.5 \times (a - t)^2$$

In the end, the weight update is defined as proportional to the input value, to the derivative of the error function for each dimension, and scaled by a learning rate* factor.

$$\omega_i \leftarrow \omega_i - \eta\ (a - t)\ x_i$$

*Typically between 0.1 and 0.4, details will be given later.

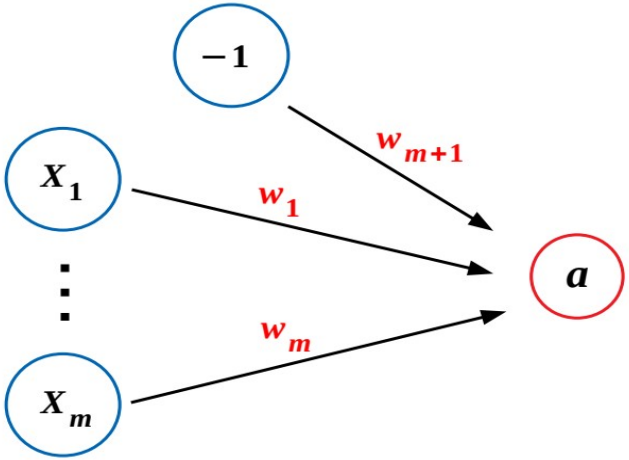
The bias node

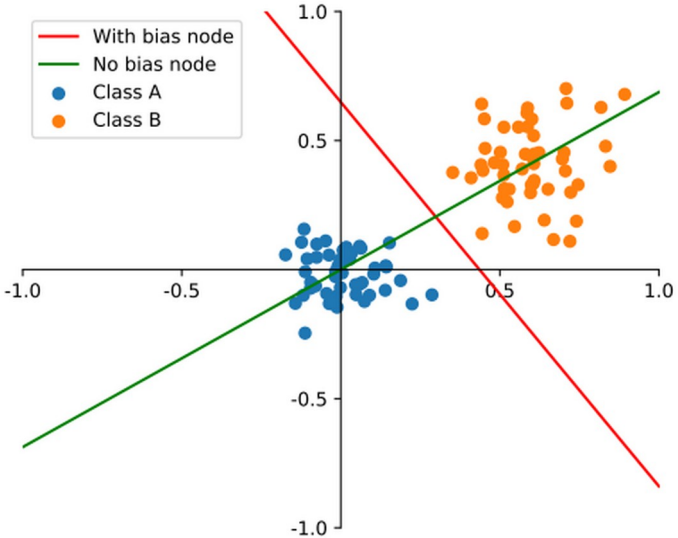
Problem with the previous formalism

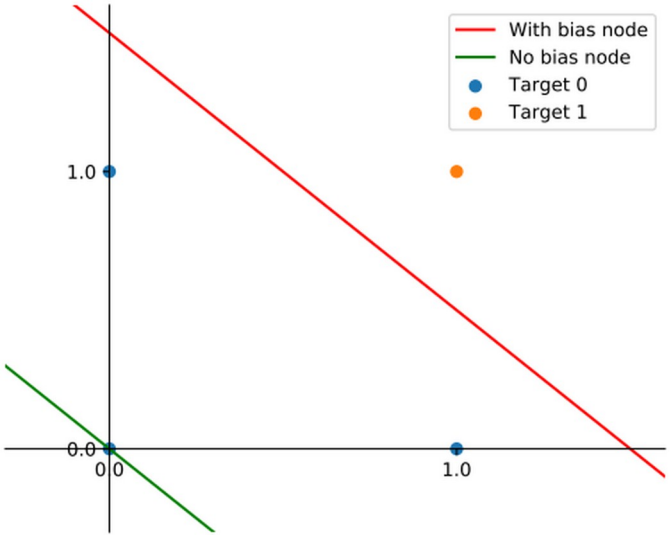
The linear separation presents a fixed *f(0) = 0 point*.
The weight correction is also 0, regardless of the input value.

Solution

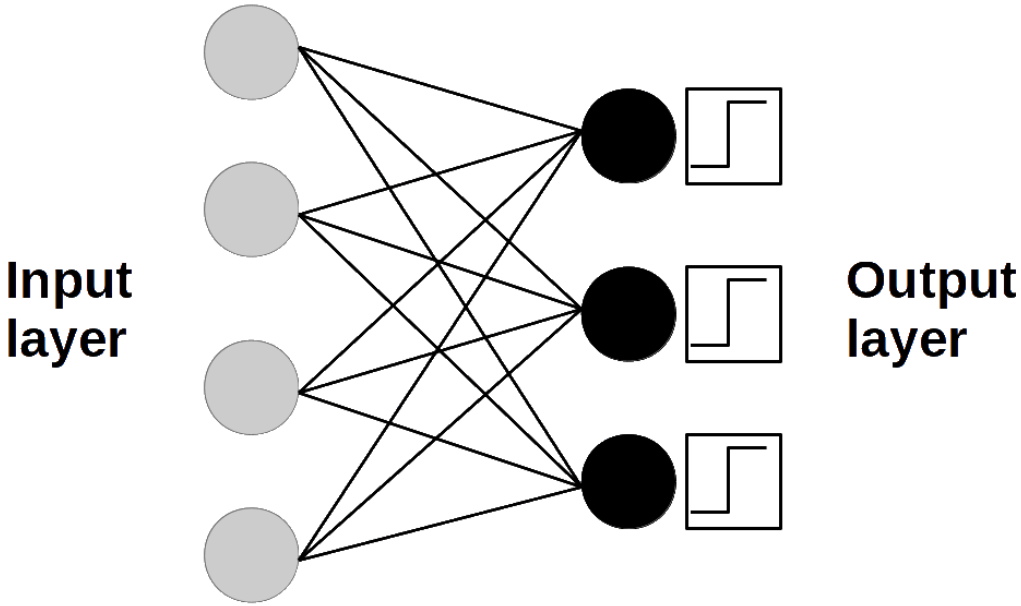
Add a bias input node that acts as an additional input with a constant value (usually -1). It has its own **variable weight** so it can learn the shift of the intercept position. The size of both the input vector and the weights vector is now *m+1*.







The Perceptron algorithm



A single neuron only performs a linear separation, which is insufficient for many applications.

The simplest way to combine neurons on a single problem is to **stack them independently**. Each neuron is connected to the input vector with its own set of weights.

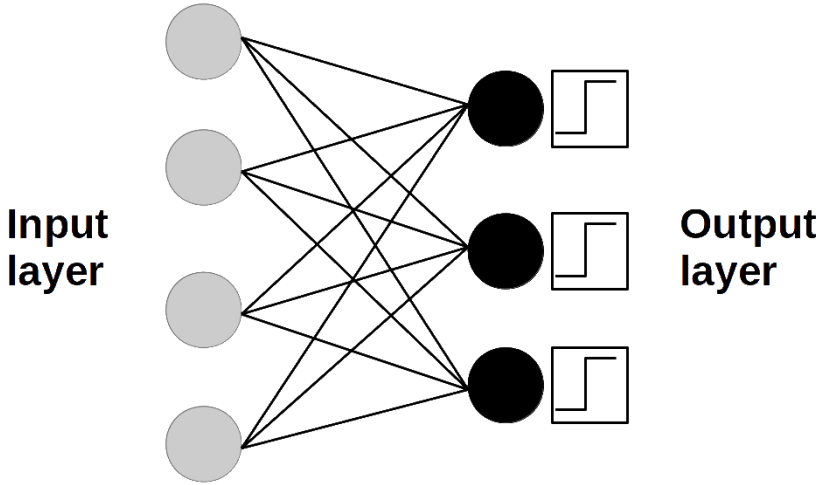
The training procedure is identical, but this time there is an index j to represent the neurons, and the weights are now in the form of a 2D matrix.

$$h_j = \sum_{i=1}^{m+1} x_i \omega_{ij} \qquad a_j = g(h_j) = \begin{cases} 1 & \text{if} \quad h_j > \theta \\ 0 & \text{if} \quad h_j \leq \theta \end{cases}$$

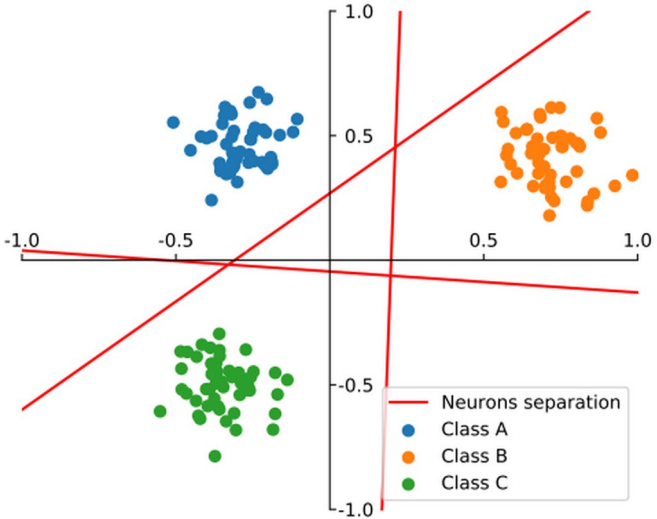
$$\omega_{ij} \leftarrow \omega_{ij} - \eta \left(a_j - t_j \right) \times x_i$$

The combination of this training procedure and this neuron connection scheme is called the single layer “Perceptron” (Rosenblatt 1958).

The Perceptron algorithm



C1	C2	C3
1	0	0
0	1	0
0	0	1



How to have multiple neurons work on the same problem ?

1) Encode the output vector (e.g in binary format)
2) Have one neuron per output class and target a format with a 1 only in the appropriate class, e.g [1,0,0]-[0,1,0]-[0,0,1]

**Warning: Since each neuron only performs a linear separation, each class must be linearly separable from all the others.*

The Perceptron algorithm

- **Initialization**
  - Set the starting weights to small random values (positive and negative). Can be drawn from a uniform or Gaussian distribution centered on zero.

- **Training**
  - For a given number of steps T, or until the output is “correct”
    - For each input vector
      - Compute the activation of each neuron *j* using the activation function *g :*

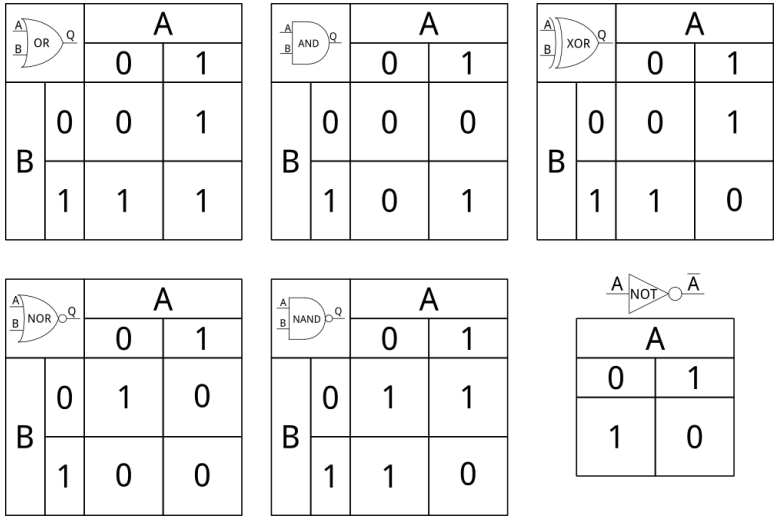
$$a_j = g\left(\sum_{i=0}^{m+1} w_{ij} x_i\right) = \begin{cases} 1 & \text{if} \quad \sum_{i=0}^{m+1} w_{ij} x_i > 0 \\ 0 & \text{if} \quad \sum_{i=0}^{m+1} w_{ij} x_i \leq 0 \end{cases}$$

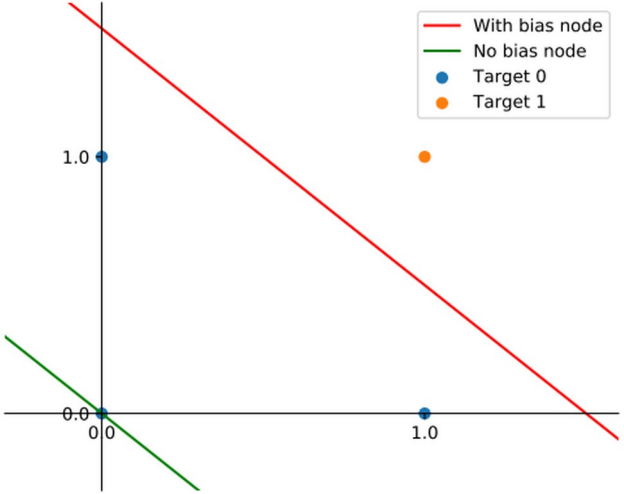
      - Update each weight individually using :

$$\omega_{ij} \leftarrow \omega_{ij} - \eta\left(a_j - t_j\right) \times x_i$$

- **Inference**
  - Compute the final activation of each neuron *j* for each input vector to test

First application : logical gates





A straightforward application of this algorithm is to make it learn a logical door, here the OR or AND gate. For this simple application, the Perceptron has a two-dimensional input vector and 4 possibilities as input-target pairs.

The output is made of a single binary neuron to predict either the 0 or 1 state of the gate.

As a first exercise:

- Write a simple program that declares an array with all the possible inputs and the associated targets.
- Identify all the necessary variables and initialize the weights with small values.
- Try to train this neuron over a few iterations following the Perceptron algorithm.

Do not forget the bias node!

- Display the output at each iteration to watch the network converge toward a stable solution.

Why is the **XOR gate** problematic with this algorithm? How to solve the problem?

The Pima Indian dataset

Dataset from the UCI Machine Learning repository.

Provides 8 physiological measurements for 768 female native Americans and provides a class depending on whether the individual has developed diabetes.

This exercise aims to implement a Perceptron to **predict if a person has diabetes** based solely on the 8 input measurements.

Despite its limitations, a simple Perceptron can reach up to **70% accuracy** on this dataset.

- Modify your program to read the Pima dataset.
- Adapt the input dimension (including bias).
- Adapt the corresponding computations.
- Find a method to compute the "accuracy" and its evolution during training.
- Question the limits of your accuracy definition, and find a way to ensure that you are testing the generalization capabilities of the method.

Attribute Number	Attribute
1	Patient age
2	Body mass index (kg/m2)
3	Concentration of plasma glucose
4	2-h serum insulin (mu U/mL)
5	Thickness of triceps skin-fold (mm)
6	Pedigree function of diabetes
7	Number of times patient pregnant
8	Diastolic blood pressure (mmHg)
9	Class 0 or 1

Input dimension : 8 (+1)

Output dimension : 1
(or 2 if using one neuron per class)

Number of example : 768

Class distribution : 500 class 0; and 268 class 1

The Iris dataset

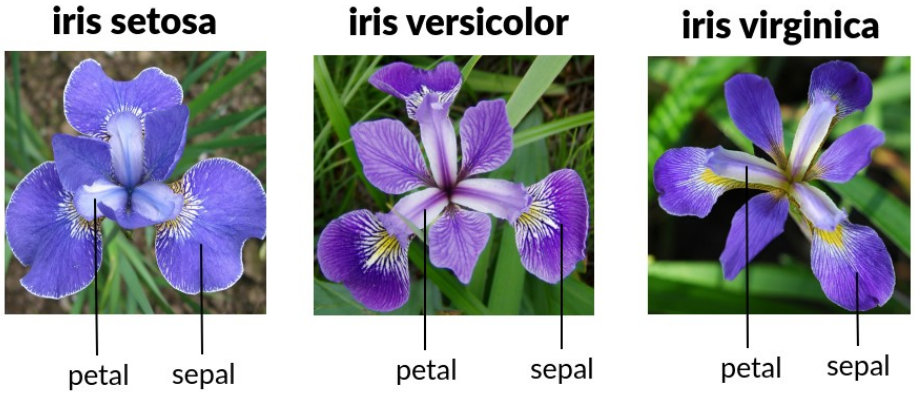
Dataset from the UCI Machine Learning repository.

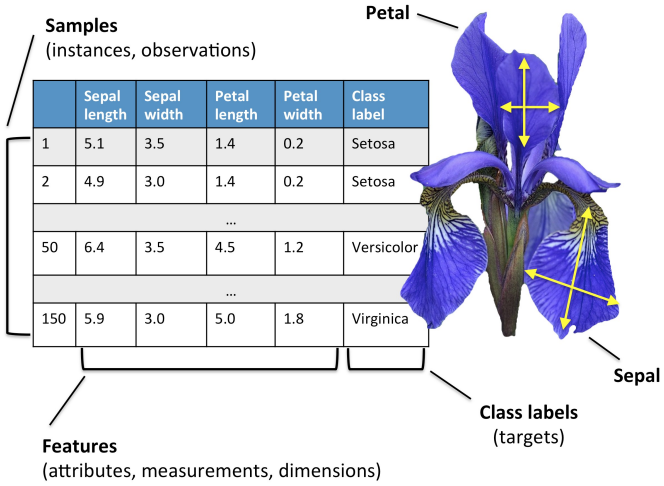
Provides 4 sizes for a set of 150 Iris flowers and provides membership to one of 3 classes.

This exercise aims to implement a Perceptron to **predict the corresponding class** based on the 4 input measurements for each flower.

Despite its limitations, a simple Perceptron can reach up to **80% accuracy** on this dataset.

- Modify your program to read the Pima dataset.
- Adapt the input dimension (including bias).
- Adapt the corresponding computations.
- Find a method to compute the "accuracy" and its evolution during the training phase.
- Question the limits of your accuracy definition, and find a way to ensure that you are testing the generalization capabilities of the method.





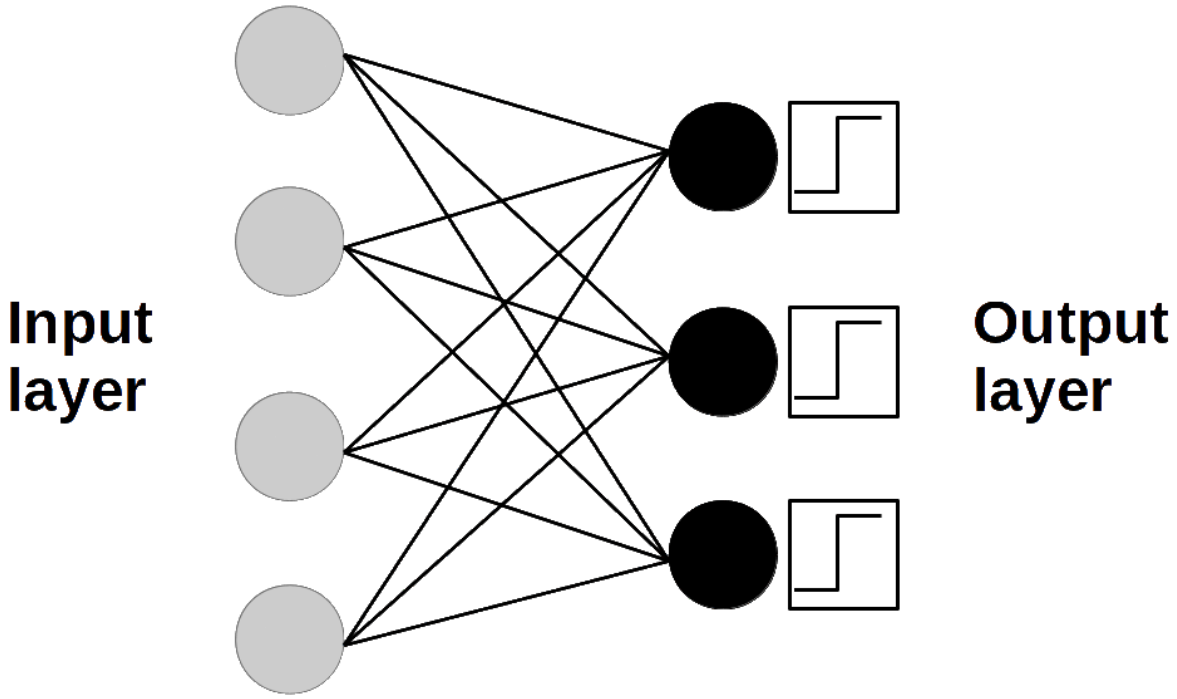
	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica

Input dimension : 4 (+1)
Output dimension : 3
Number of example : 150
Class distribution : 50 examples per class

Constructing a “Deep” Neural Network

For more complex problems more neurons must be used.

First: stacked independently in a layer (Perceptron)



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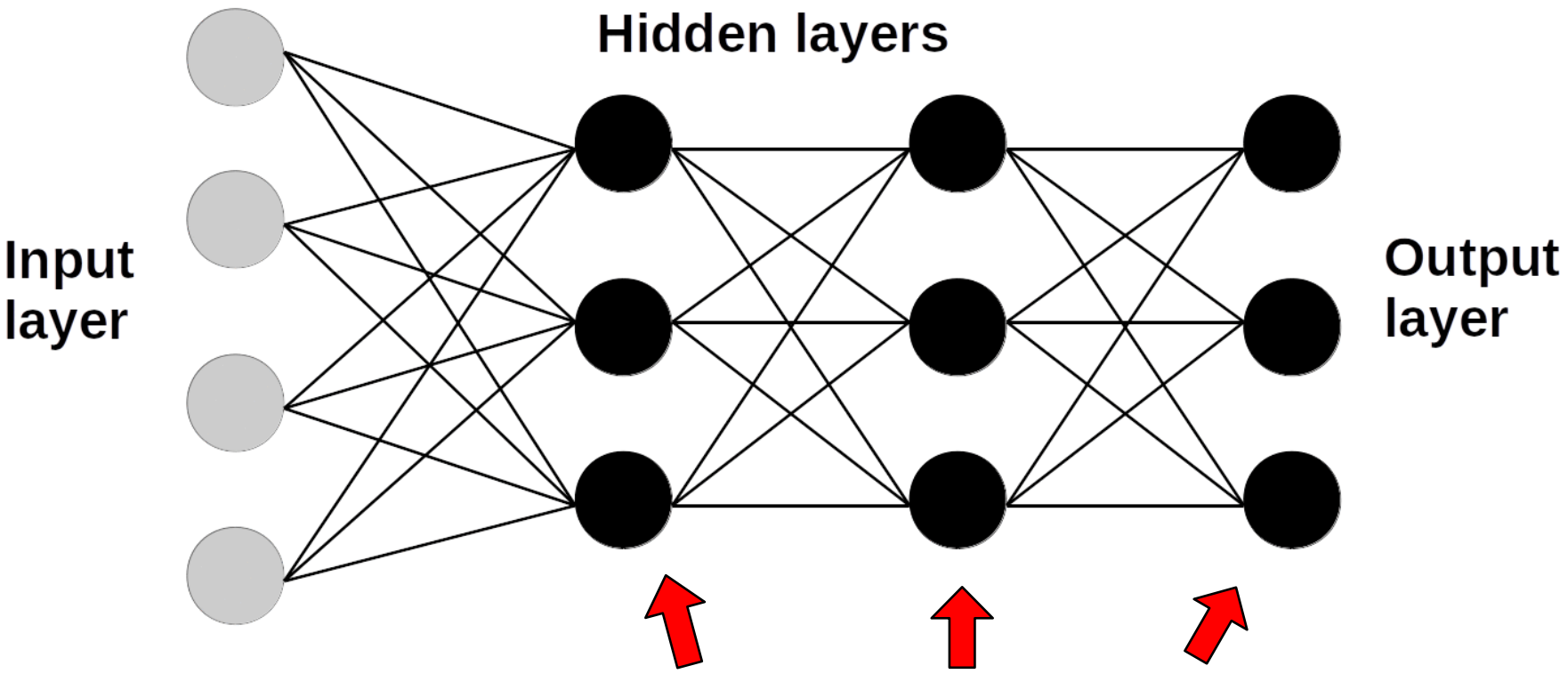


Then: stack several layers (Multi Layer Perceptron)

Constructing a “Deep” Neural Network

In addition to the MLP architecture, it is necessary to change the activation to allow non-linear combinations

Change for the Sigmoid (logistic) function

$$g(h) = \frac{1}{1 + \exp(-\beta h)}$$

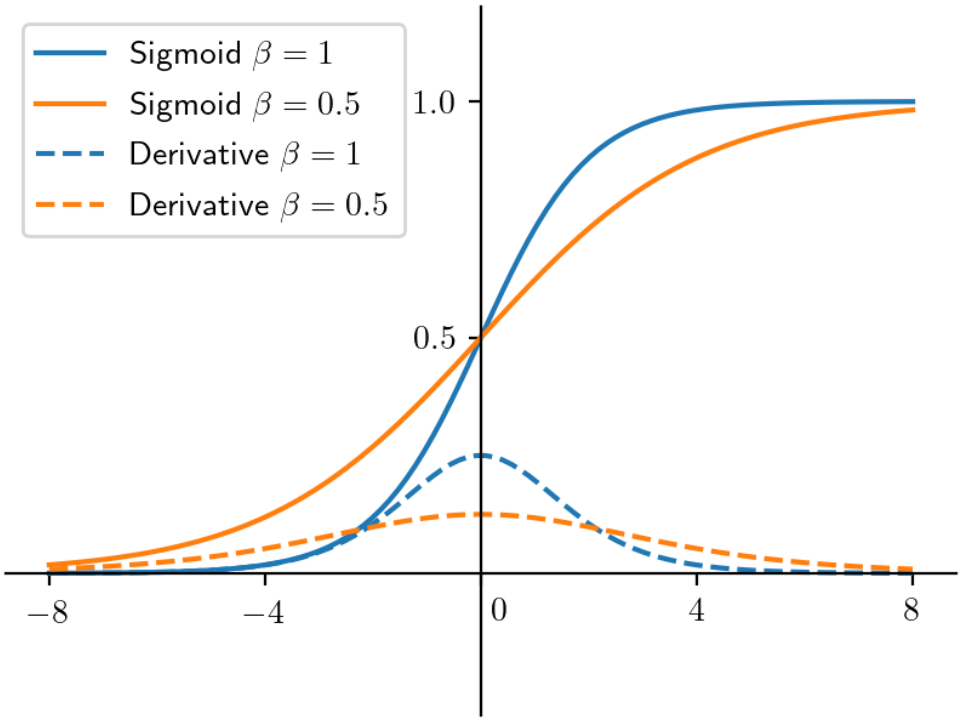
Beta represents the slope of the function.

The sigmoïd is continuous while preserving the desired global binary behavior in its extremity.

It has a nice derivative form :

$$\frac{\partial a_j}{\partial h_j} = \beta a_j (1 - a_j)$$

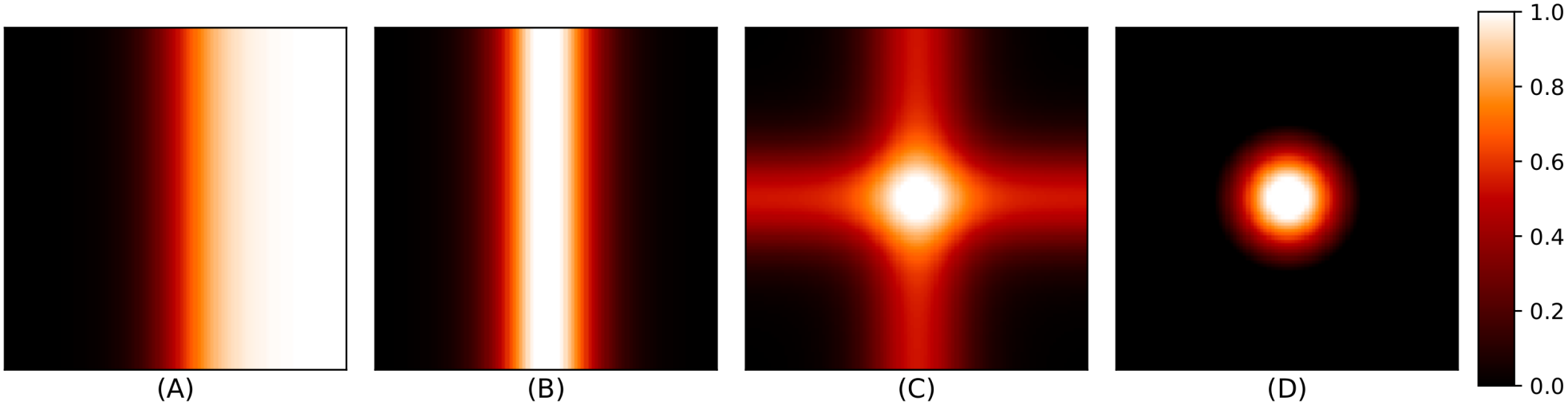
For regression problems, the output layer can either use sigmoïds or simple linear activation functions since the previous layers will handle non-linearity.



Constructing a “Deep” Neural Network

Stacking layers that use a sigmoid activation allows the construction of more and more complex functions. The image below demonstrates that it is possible to reconstruct hill shapes (B), bumps (C), and point-like functions.

➡ With these improvements, the MLP is a **Universal Function Approximator**



In practice, based on the universal approximation theorem, a MLP with a single hidden layer remains a UFA at the condition of having a large enough number of neurons.

Example with classification

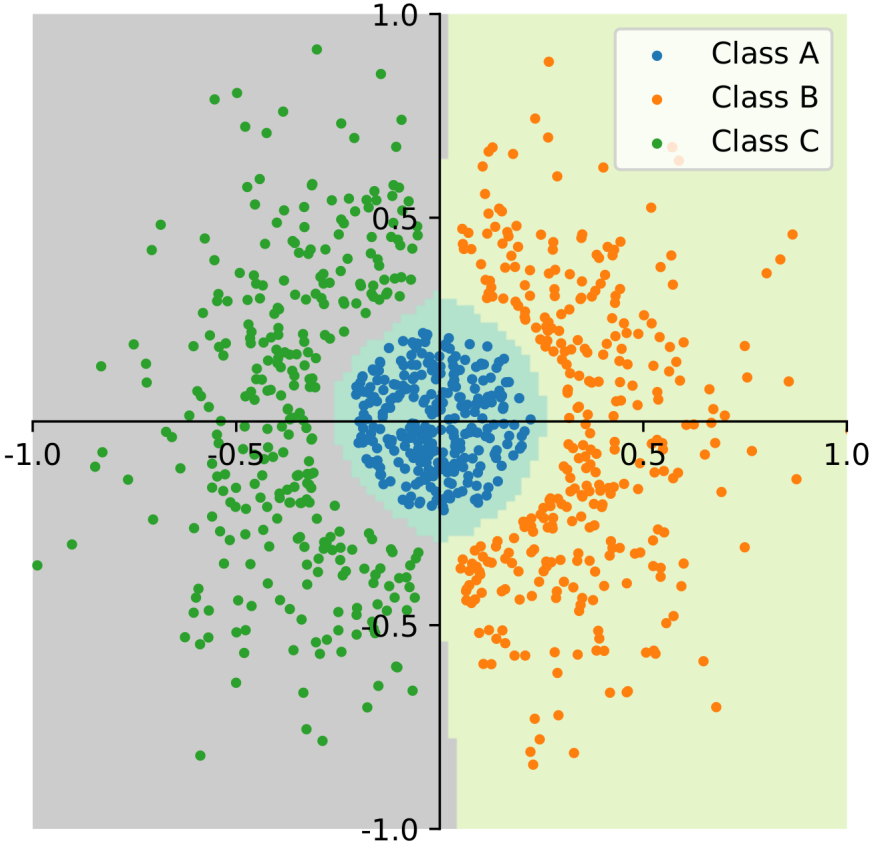
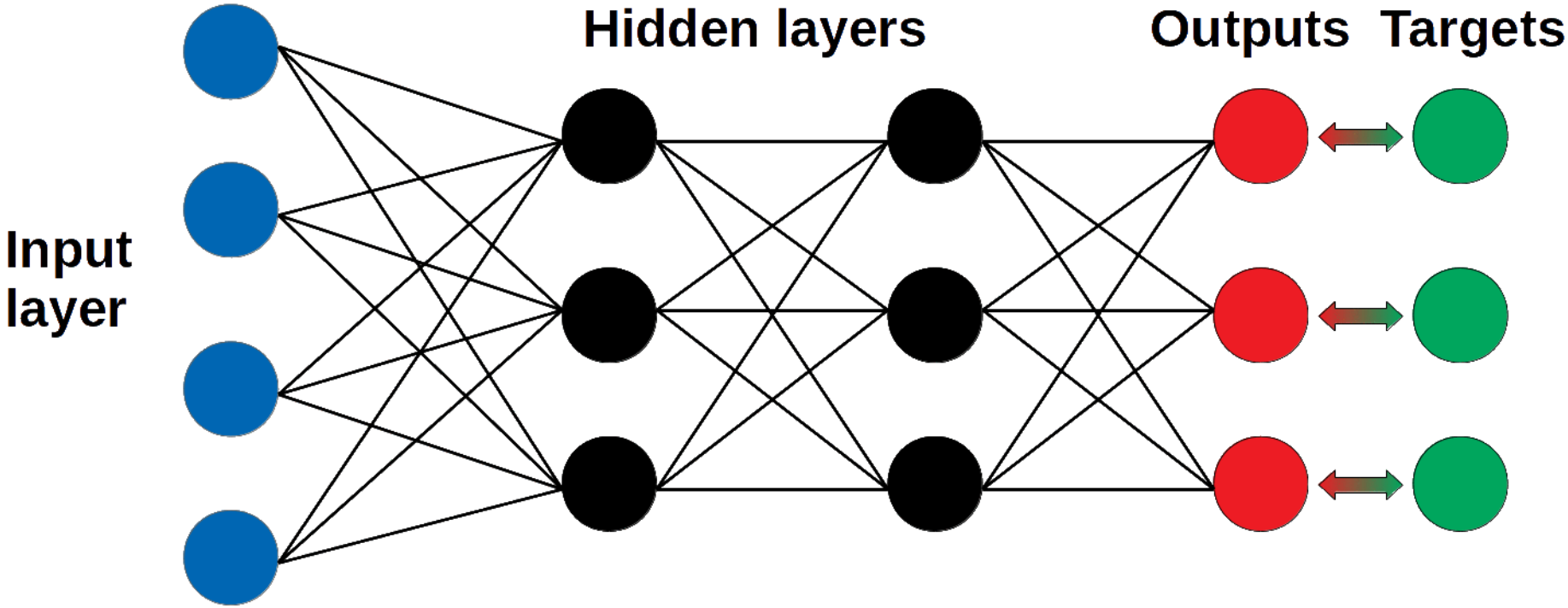


Illustration of a three-class separation in a two-dimensional feature space using a trained MLP with 3 output neurons and 8 hidden neurons, all with sigmoid activations. The light background colors indicate the regions of the feature space that the network has attributed to each class.

Error gradient backpropagation

Problem : how to correct the output of neurons in the hidden layers ?



$$\omega_{ij} \leftarrow \omega_{ij} - \eta \frac{\partial E}{\partial \omega_{ij}} \quad \Rightarrow \quad \frac{\partial E}{\partial \omega_{ij}} = \delta_l(j) \frac{\partial h_j}{\partial \omega_{ij}} \quad \text{with} \quad \delta_l(j) \equiv \frac{\partial E}{\partial h_j} = \frac{\partial E}{\partial a_j} \frac{\partial a_j}{\partial h_j} = \frac{\partial a_j}{\partial h_j} \sum_k \omega_{kj} \delta_{l+1}(k)$$

Actual Network equations

Changes in the Perceptron algorithm required to obtain the Multi-Layer Perceptron algorithm :

- Change the activation function of all neurons, with $\beta \geq 1$

$$g(h) = \frac{1}{1 + \exp(-\beta h)}$$

- Add a hidden layer with its own bias node (connect input → hidden, and hidden → output)
- Implement the "back-propagation" with $E(a^o, t) = \frac{1}{2} \sum_{k=1}^{N} (a_k^o - t_k)^2$ using the following equations:

$$\delta_o(k) = \beta a_k^o (1 - a_k^o)(a_k^o - t_k)$$

$$\delta_h(j) = \beta a_j^h (1 - a_j^h) \sum_{k=1}^{o} \delta_o(k) \omega_{jk}$$

$$\omega_{jk} \leftarrow \omega_{jk} - \eta \delta_o(k) a_j^h$$

$$v_{ij} \leftarrow v_{ij} - \eta \delta_h(j) x_i$$

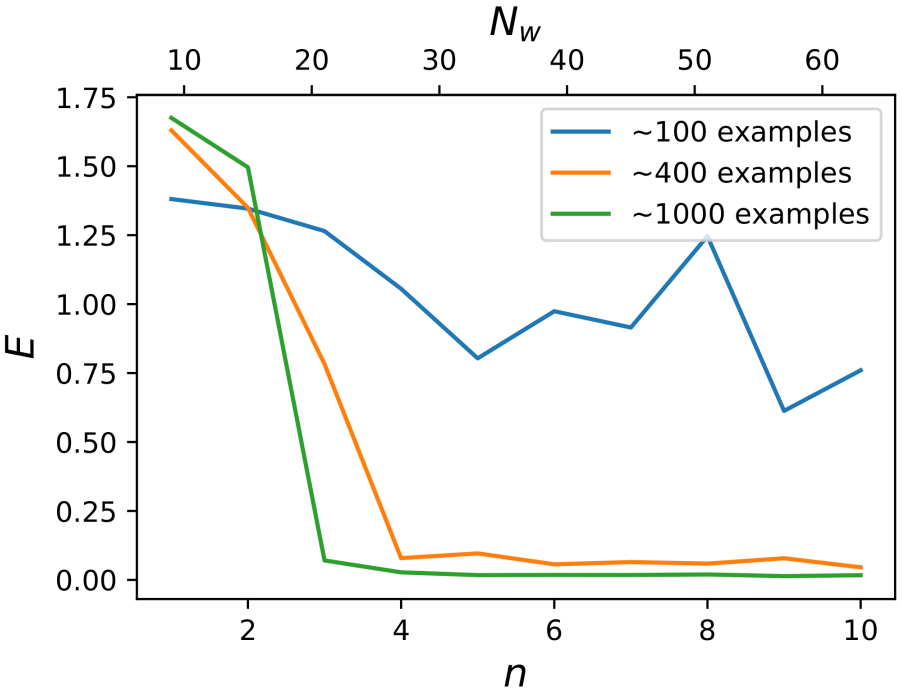
Network depth and size

How to choose the network architecture?
In the case of the MLP, how many layers and neurons should be used per layer?

In this context, the objective is to find a **trade-off between:**
The strength of the network (its expressive power) and the difficulty of training the model

A few ideas that help explore the architectures:

- More data allows to constrain more parameters
  Rule of thumb: need roughly 10 times the number of weights as training examples
- More neurons increase the strength but add parameters to fit.
  Reminder: a neuron is a linear separation in the parameter space → how many would you need by eye? (starting point to explore)
- More layers allow the construction of complex functions with fewer parameters, but increases the difficulty.
  Reminder: a single hidden layer can approximate any function, but it is not always the optimal solution (nb. of parameters, time to converge, etc.)



Finding the appropriate architecture is always an exploratory and iterative process.
Training different networks on the same data is almost always necessary to obtain good results.

Learning rate

Choosing the appropriate learning rate for the task is very important and must consider several aspects of the error space.

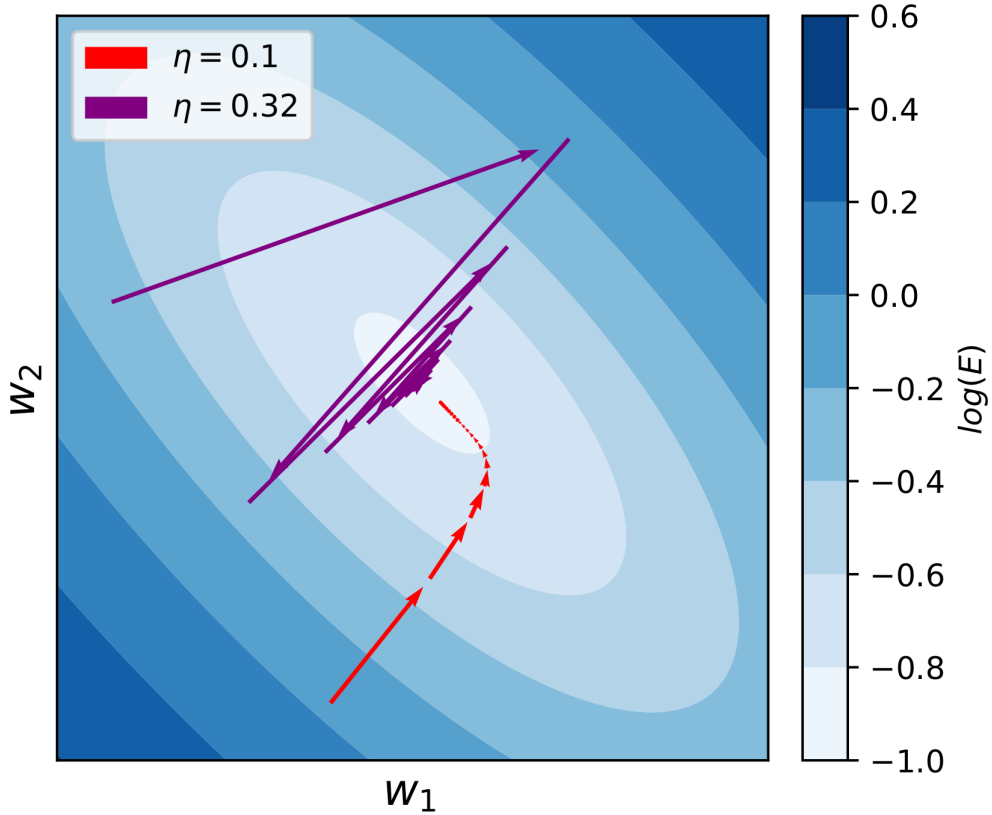
	Small η	**Large η**
Pros	• Increase training stability • Can find narrower minima	• Faster training • Easily switch from a local minima to another
Cons	• Slow training • Might get stuck on local minima	• Might fail to find a narrow minima • Training can be very unstable

The learning rate does not have to be a single value, it can change during the training phase.

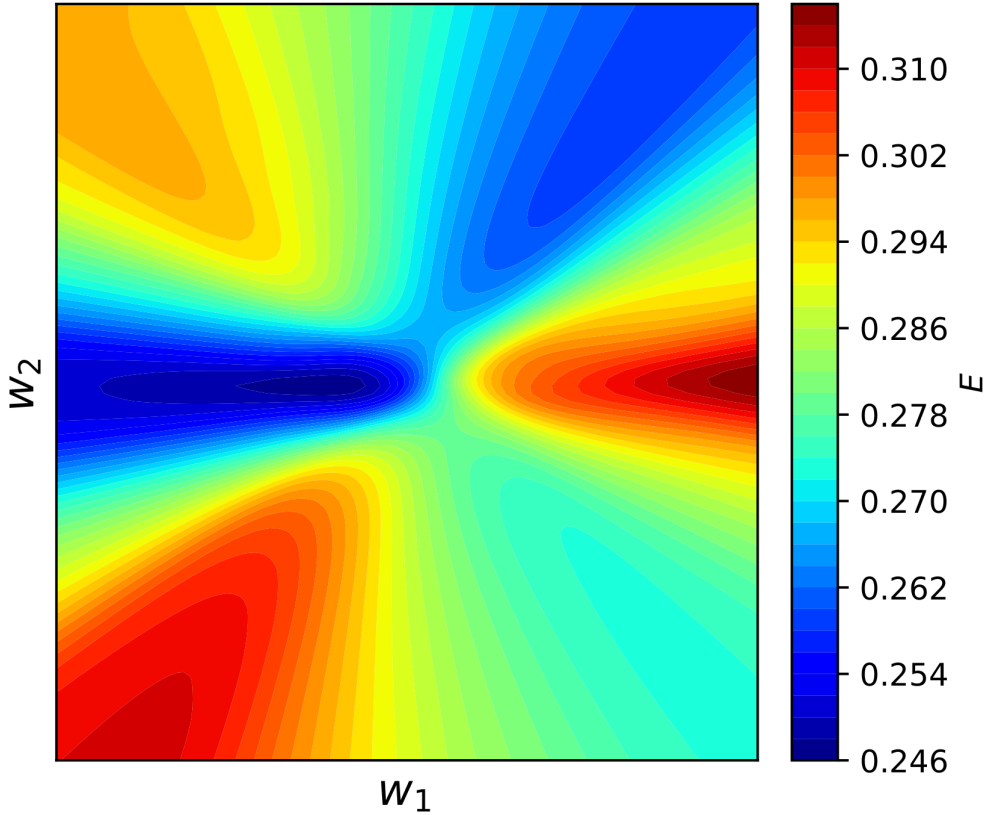
The simple approaches are: pre-established fixed values, linear decay, exponential decay, ...

Most frameworks allow the use of advanced gradient descent scheme that also automatically adapts the learning rate: Adam, RMSprop, Adagrad, etc ...

Learning rate



Logarithm of the error term in the weight space of a simple binary neuron with a two dimensional input parameter. The arrows represent the correction after each epoch for two different learning rates.



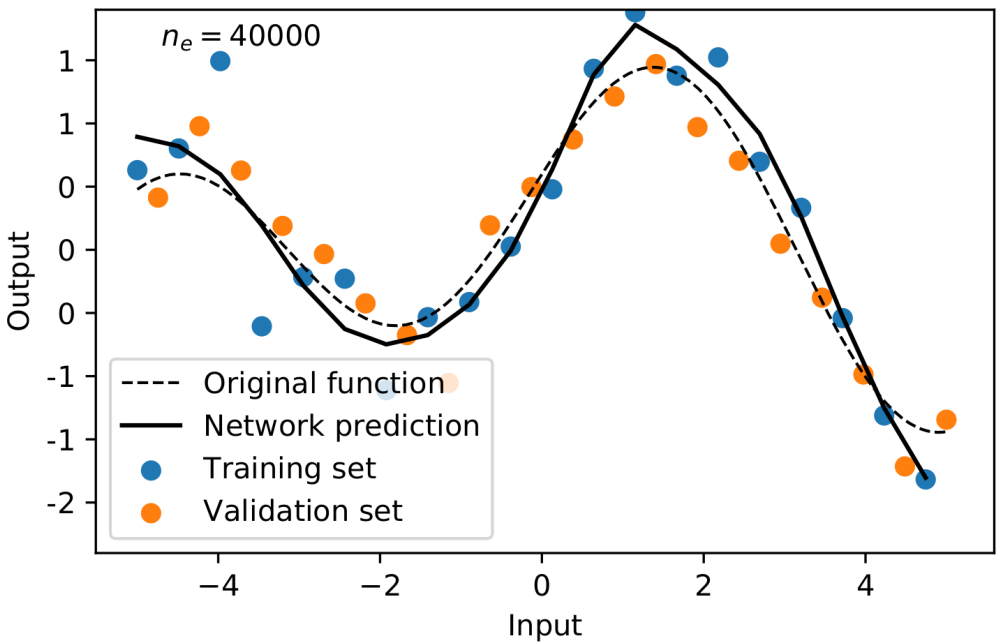
Error term in a 2D weight space of neuron in the hidden layer of a 3 class output MLP. The other weights of the network are frozen while exploring this 2D weight space.

Overtraining

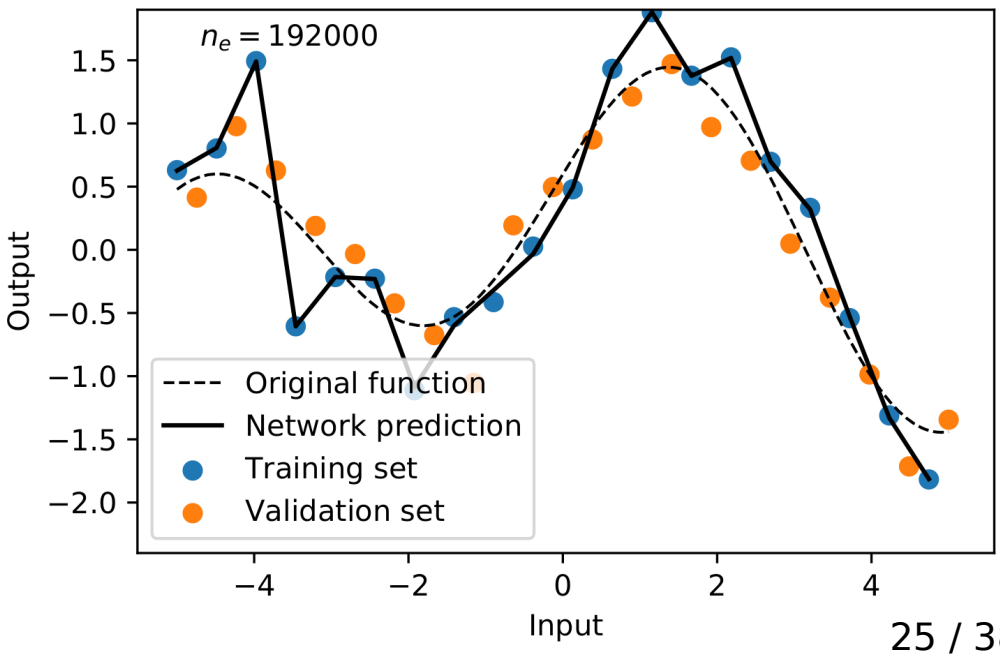
Overtraining: At some point during the training phase, the network that looks at the same data again and again, might start to over-fit!

It means that the network is learning specificities of the dataset that are no longer related to the average function that it should approximate.

Proper fit



Over-fitting

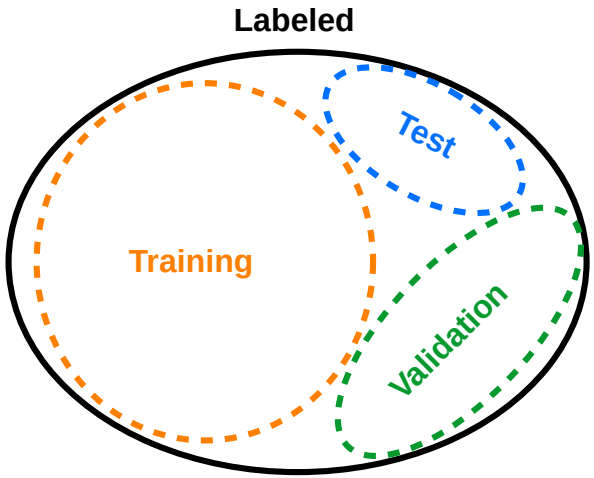


Monitoring overtraining

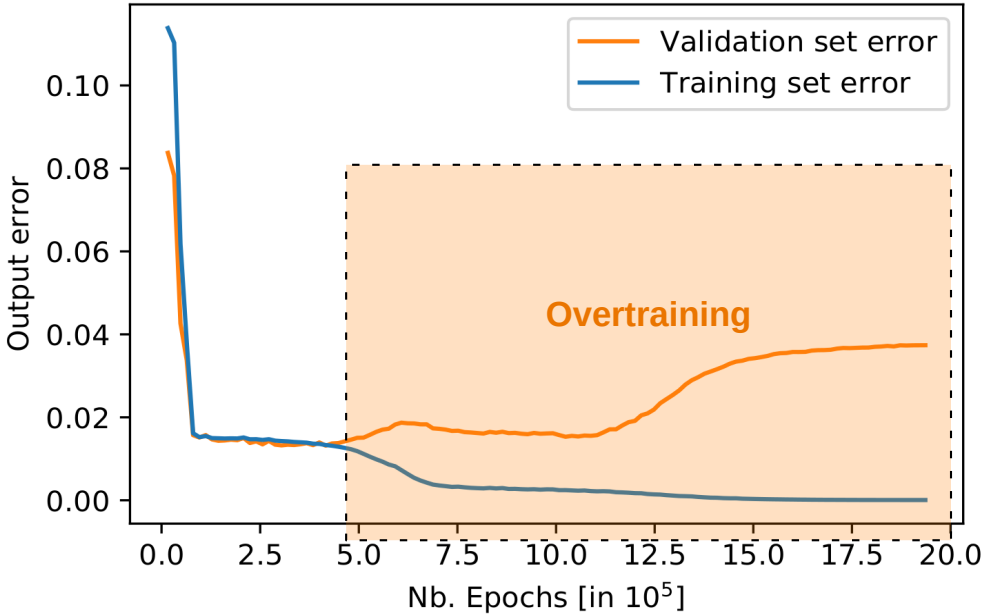
While several techniques can be used to moderate overtraining (like regularization), it is always necessary to monitor it to stop the training at the appropriate time.

Split **labeled dataset** into 3 subset

- **Training set** ➤ To train the network
- **Validation set** ➤ To monitor the training process
- **Test set** ➤ To assess the quality of the prediction



Typical proportions



Improving results with data pre-processing

Improving the prediction results relies on mode than just using larger networks.
Data pre-processing can be employed with two primary objectives :

Add already known information, remove irrelevant information

The way we organize and process the data can help statistical methods.

For example, there is an “age” parameter in the Pima dataset. While it is well known that most medical issues correlate to aging, a one-year resolution is unlikely to reflect the effect. Using groups based on the age parameter, e.g. [Under 30, 30-40, 40-50, 50+], tends to improve the result.

Another example is the “number of times pregnant”. While there is certainly a significant difference between 0 and 1, it is unlikely that 3+ makes more difference. So it should be possible to cap this value.

This type of pre-processing can have a significant impact on prediction performance for large models.

Balance the numerical information

Represents any modification that allows a better balance in the algorithm itself. For example, it is almost always necessary to **normalize each dimension of the dataset** in the range [0,1]. Otherwise, some dimensions might dominate the weight update gradient, eclipsing the information contained in the others.

Many hyper-parameter choices (normalization range, weight init. range, learning rate, etc.) are made to improve the learning process balance, which is often critical to constructing models that work at all.

Focus on normalization

PIMA attributes statistics
→ each dimension is scaled differently

Attribute number:	Mean:	Standard Deviation:
1.	3.8	3.4
2.	120.9	32.0
3.	69.1	19.4
4.	20.5	16.0
5.	79.8	115.2
6.	32.0	7.9
7.	0.5	0.3
8.	33.2	11.8

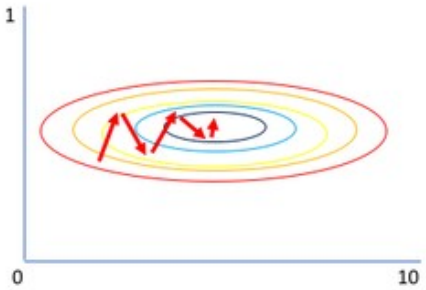
Different approaches to rescaling

Centered: $X - \text{mean}$

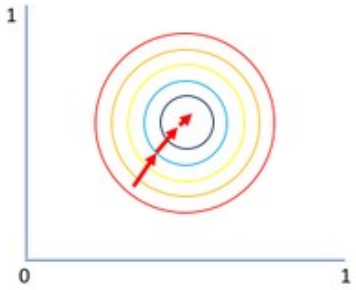
Standardized: $\dfrac{X - \text{mean}}{\text{sd}}$

Normalized: $\dfrac{X - \min(X)}{\max(X) - \min(X)}$

Why normalize ?



Gradient of larger parameter dominates the update



Both parameters can be updated in equal proportions

WARNING !

All labeled data must be normalized with the same values (not only with the same equation)!

This include both the training and testing samples but also the data used for prediction at deployment time!

How to measure classification performance ?

Confusion Matrix

- Uses "**observational proportions**"
- Provides confusion between classes
- Is much more precise than the average accuracy

$$\text{Recall} = \frac{TP}{TP + FN}$$

Predicted

Actual

Class	Positive	Negative	Recall
Positive	93	7	93.0%
Negative	46	954	95.4%
Precision	66.9%	99.27%	95.18%

$$\text{Precision} = \frac{TP}{TP + FP}$$

$TP \equiv$ True Positive $\quad TN \equiv$ True Negative

$FP \equiv$ False Positive $\quad FN \equiv$ False Negative

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

ML methods usually learn better on balanced dataset **BUT**
the results must always be represented using "observational proportions", with imbalance.

MLP application

Use the previous list of changes to implement the Multi-Layer Perceptron algorithm.

Use the previous datasets (iris and pima) to test your algorithm, and try to optimize its prediction performance by tweaking the dataset and the accessible network hyperparameters.

Your implementation must have the following properties :

- Easy change of the learning rate, number of epochs, number of neurons in the hidden layer
- An output encoding using one neuron per class
- A display of the results in the form of a confusion matrix

In addition, you can try to have:

- Generic functions for the forward and backprop operations that are independent of the dataset
- A way to re-balance the proportions of classes for the training phase (bet test on true proportions)

Advanced

- Splitting of the labeled data to avoid overtraining
- A probabilistic prediction (softmax or other)
- An alternative gradient descent scheme (either SGD or Batch)
- A matrix formalism version of the forward and backprop operations